

Didactic Transposition and AI Literacy

Seminar 1 • Playful AI with IoT and Generative AI

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This morning you already did the theory

You sorted cards and argued about thermostats.

You read raw sensor data and guessed at cupboards.

Both activities were transpositions: research-level ideas, rebuilt into something you could argue about before coffee.

This hour names what you were doing.

Knowledge does not travel unchanged

The core claim (Chevallard, 1985)

Knowledge is transformed every time it moves:

Scholarly knowledge

what researchers produce



Knowledge to be taught

curricula, textbooks, exam requirements



Taught knowledge

what happens in your classroom



Learned knowledge

what students actually take away

Each arrow is a transformation, not a transfer.

Every arrow changes the knowledge

At each step, something is:

- **Selected** — most of the field is left out
- **Simplified** — continuous things become discrete, messy things become clean
- **Recontextualised** — removed from research problems, attached to school problems
- **Time-structured** — split into lessons, sequenced, made examinable

None of this is corruption. **It is the condition for teachability.**

The question is never *whether* to transform, but whether the transformation is honest.

Why AI makes this urgent

The AI transposition chain is unusually broken:

- Scholarly knowledge moves **fast** — the textbook step cannot keep up
- The public chain runs through **media and marketing**, not curricula
- The dominant metaphors — *thinking, knowing, hallucinating* — are borrowed from minds, and they mislead
- Your students arrive **already taught**, by TikTok, by product launches, by each other

You are not writing on a blank slate. You are correcting a transposition already done badly.

**Watch the transposition in this
morning's activities**

"Is this AI?" — transposing a definition problem

Scholarly version: decades of unresolved debate about intelligence, agency, and learning in the research literature.

This morning's version: 36 cards and an argument.

What survived: the genuine instability of the concept.

What was cut: the entire technical apparatus.

The honest move: the activity never pretends the question has an answer. The disagreement *is* the content.

Sensor blind tasting — transposing "perception"

Scholarly version: signal processing, feature extraction, the symbol grounding problem.

This morning's version: a number on a screen, a guess about a cupboard.

The gap you felt — between *light level: 12* and *"it's in the dark cupboard"* — is the gap the whole field lives in.

The honest move: the activity lets you feel the gap instead of defining it.

The dishonest transposition, for contrast

Familiar from textbooks and news media:

- "The AI **understands** your question"
- "The model **knows** billions of facts"
- A brain image with glowing circuits
- The robot hand touching the human hand

Each one transposes AI into folk psychology.

Fluent, memorable — and it points students' intuitions in exactly the wrong direction.

A working tool, not just a theory

The transposition check

Three questions to ask of any AI activity — yours or anyone's:

1. **What was simplified?**

Name the cuts explicitly. There are always cuts.

2. **Is the simplification honest?**

Does it point intuition in roughly the right direction, or does it plant a misconception that must later be unlearned?

3. **What could a student wrongly conclude?**

The best activities know their own failure modes.

You will apply this check to everything you build on Thursday.

AI literacy, restated

On this view, AI literacy is **not**:

- a list of facts about neural networks
- tool training for the current interface
- pure critique from a safe distance

It **is** the capacity to ask, of any AI claim or system:

what is actually going on here, what has been simplified for me, and by whom?

That capacity is built through honest transpositions — which is what this workshop is for.

Where this goes next

- **This afternoon:** you build a paper classifier and watch it fail. Ask the three questions of it.
- **Tomorrow:** the same lens on generative AI — where the transposition problem gets harder.
- **Thursday:** you design your own activity. The transposition check is the design brief.
- **October:** you find out what your classroom did to your transposition.

Reading, if you want it

Achiam, didactic transposition as a research programme
(link in the programme document)

The wall definitions from this morning stay up all five days.
Revise them as you go.